29/01/2024
Evening

*Corporate Office* : Aakash Tower, 8, Pusa Road, New Delhi-110005 | Ph.: 011-47623456

# Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

# JEE (Main)-2024 (Online) Phase-1

## (Mathematics, Physics and Chemistry)

**IMPORTANT INSTRUCTIONS:**

(1) The test is of **3 hours** duration.

(2) This test paper consists of 90 questions. Each subject (MPC) has 30 questions. The maximum marks are 300.

(3) This question paper contains **Three Parts**. **Part-A** is Mathematics, **Part-B** is Physics and **Part-C** is **Chemistry**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt any 05 questions out of 10 Questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 30) contains** 10 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



# MATHEMATICS

## SECTION - A

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer:**

1. If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then $\alpha^2$ is equal to

(1) 4 (2) 9

(3) 3 (4) 12

**Answer (3)**

**Sol.** $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} - y\cos\left(\frac{y}{x}\right) = x$

$\Rightarrow \cos\left(\frac{y}{x}\right)\left[x\frac{dy}{dx} - y\right] = x$

$\Rightarrow \cos\left(\frac{y}{x}\right)\left[\frac{x\frac{dy}{dx} - y}{x^2}\right] = \frac{1}{x}$

Let $\frac{y}{x} = t$

$\frac{x\frac{dy}{dx} - y}{x^2} = \frac{dt}{dx}$

$\Rightarrow \cos t\frac{dt}{dx} = \frac{1}{x}$

$\Rightarrow \int \cos t\, dt = \int \frac{1}{x}dx$

$\Rightarrow \sin t = \ln|x| + C$

$y(1) = \frac{\pi}{3}$

$\Rightarrow \sin\frac{\pi}{3} = 0 + C \Rightarrow C = \frac{\sqrt{3}}{2}$

$\therefore$ Solution is $\sin\frac{y}{x} = \ln|x| + \frac{\sqrt{3}}{2}$

$\therefore$ $\alpha^2 = 3$

2. The function $f(x) = \frac{x}{x^2 - 6x - 16}, x \in \mathbb{R} - \{-2, 8\}$

(1) Decreases in $(-2, 8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

(2) Decreases $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(3) Increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(4) Decreases $(-\infty, -2)$ and increases in $(8, \infty)$

**Answer (2)**

**Sol.** $f(x) = \frac{x}{x^2 - 6x - 16}$

$f'(x) = \frac{(x^2 - 6x - 16) - x(2x - 6)}{(x^2 - 6x - 16)^2}$

$= \frac{x^2 - 6x - 16 - 2x^2 + 6x}{(x^2 - 6x - 16)^2}$

$= \frac{-x^2 - 16}{(x^2 - 6x - 16)^2} < 0 \ \forall \ x \in D_f$

$\therefore$ $f(x)$ is decreasing in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

3. Let $P(3, 2, 3)$, $Q(4, 6, 2)$ and $R(7, 3, 2)$ be the vertices of $\Delta PQR$. Then, the angle $\angle QPR$ is

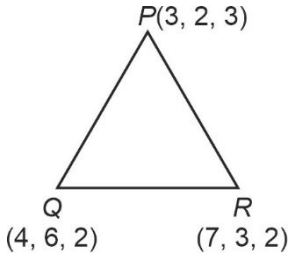
(1) $\frac{\pi}{6}$ (2) $\cos^{-1}\left(\frac{7}{18}\right)$

(3) $\frac{\pi}{3}$ (4) $\cos^{-1}\left(\frac{1}{18}\right)$

**Answer (3)**



**Sol.** $P(3, 2, 3)$, $Q(4, 6, 2)$, $R(7, 3, 2)$




Direction ratio of $PR$ = 4, 1, 1

Direction ratio of $PQ$ = 1, 4, 1

$$\cos(\angle QPR) = \left|\frac{1\times 4 + 4\times 1 + 1\times 1}{\sqrt{16+1+1}\sqrt{1+16+1}}\right|$$

$$\cos(\angle QPR) = \frac{1}{2}$$

$$\Rightarrow \quad \angle QPR = \cos^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow \quad \angle QPR = \frac{\pi}{3}$$

4. Let $x = \frac{m}{n}$ ($m$, $n$ are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1} x\right) = \frac{1}{9}$ and let $\alpha, \beta$ ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point $(\alpha, \beta)$ lies on the line

(1) $5x + 8y = 9$ (2) $5x - 8y = -9$

(3) $3x + 2y = 2$ (4) $3x - 2y = -2$

**Answer (1)**

**Sol.** $\cos\left(2\sin^{-1} x\right) = \frac{1}{9}, \; x > 0$

$\because \quad x > 0$

$\sin^{-1}x = \theta \in 1^{st}$ Quadrant

$$\Rightarrow \quad \cos 2\theta = \frac{1}{9}$$

$$= 1 - 2\sin^2\theta = \frac{1}{9}$$

$$\Rightarrow \quad \sin^2\theta = \frac{4}{9}$$

$$\Rightarrow \quad x^2 = \frac{4}{9}$$

$$\Rightarrow \quad x = \frac{2}{3} = \frac{m}{n}$$

$\Rightarrow \quad m = 2$ and $n = 3$

$mx^2 - nx - m + n = 0$

$\Rightarrow \quad 2x^2 - 3x + 1 = 0$

Roots are $1, \frac{1}{2}$

$$\alpha = 1, \; \beta = \frac{1}{2}$$

So, $(\alpha, \beta) = \left(1, \frac{1}{2}\right)$

and $(\alpha, \beta)$ lies on $5x + 8y = 9$

Hence, option (1) is correct

5. The sum of the solution $x \in \mathbb{R}$ of the equation $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is

(1) 3 (2) 1

(3) –1 (4) 0

**Answer (3)**

**Sol.** Given: $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$

$$\Rightarrow \frac{\cos 2x(3 + \cos^2 2x)}{(\cos^2 x - \sin^2 x)[\sin^4 x + \cos^4 x + \sin^2 x \cos^2 x]} = x^3 - x^2 + 6$$

$$\Rightarrow \quad \frac{3 + \cos^2 2x}{1 - \sin^2 x \cos^2 x} = x^3 - x^2 + 6$$

$$\Rightarrow \quad 4\left(\frac{3 + \cos^2 2x}{4 - \sin^2 2x}\right) = x^3 - x^2 + 6$$

$\Rightarrow \quad x^3 - x^2 + 2 = 0$

$\Rightarrow \quad (x + 1)(x^2 - 2x + 2) = 0$

$\Rightarrow \quad (x + 1)((x - 1)^2 + 1) = 0$

$\therefore$ Sum of solutions = –1



6. Let $r$ and $\theta$ respectively be the modulus and amplitude of the complex number $z = 2 - i\left(2\tan\frac{5\pi}{8}\right)$, then $(r, \theta)$ is equal to

(1) $\left(2\sec\frac{3\pi}{8}, \frac{5\pi}{8}\right)$ (2) $\left(2\sec\frac{5\pi}{8}, \frac{3\pi}{8}\right)$

(3) $\left(2\sec\frac{11\pi}{8}, \frac{11\pi}{8}\right)$ (4) $\left(2\sec\frac{3\pi}{8}, \frac{3\pi}{8}\right)$

**Answer (4)**

**Sol.** $z = 2 - 2i\dfrac{\sin\frac{5\pi}{8}}{\cos\frac{5\pi}{8}}$

$= \dfrac{2}{\cos\frac{5\pi}{8}}\left(\cos\frac{5\pi}{8} - i\sin\frac{5\pi}{8}\right)$

$= 2\sec\frac{5\pi}{8}e^{i\left(-\frac{5\pi}{8}\right)}$

$= 2\sec\frac{3\pi}{8}e^{i\pi}e^{i\left(-\frac{5\pi}{8}\right)}$

$= 2\sec\frac{3\pi}{8}e^{i\frac{3\pi}{8}}$

$\therefore \quad r = 2\sec\frac{3\pi}{8},\ \theta = \frac{3\pi}{8}$

7. Let $A$ be the point of intersection of the lines $3x + 2y = 14$, $5x - y = 6$ and $B$ be the point of intersection of the lines $4x + 3y = 8$, $6x + y = 5$. The distance of the point $P(5, -2)$ from the line $AB$ is

(1) $\frac{5}{2}$ (2) 8

(3) 6 (4) $\frac{13}{2}$

**Answer (3)**

**Sol.** $A$ is point of intersection of line $3x + 2y = 14$ and $5x - y = 6$

On solving, $3x + 2y = 14$

$10x - 2y = 12$

$13x = 26$

$x = 2$ and $y = 4$

Point $A(2, 4)$

Similarly, $B$ is $4x + 3y = 8$

$18x + 3y = 15$

$14x = 7$

$x = \frac{1}{2}$ and $y = 2$

Point $B\left(\frac{1}{2}, 2\right)$

$\therefore$ Line $AB$ is $y - 4 = \dfrac{-2}{-\frac{3}{2}}(x - 2)$

$y - 4 = \frac{4}{3}(x - 2)$

$3y - 12 = 4x - 8$

$4x - 3y + 4 = 0$

$\therefore$ Distance of $P(5, -2)$ from $AB$ is

$= \dfrac{|4(5) - 3(-2) + 4|}{\sqrt{(4)^2 + (-3)^2}}$

$= \dfrac{|20 + 6 + 4|}{5}$

$= 6$

8. An integer is chosen at random from the integers 1, 2, 3, …, 50. The probability that the chosen integer is a multiple of atleast one of 4, 6 and 7 is

(1) $\frac{14}{25}$ (2) $\frac{8}{25}$

(3) $\frac{21}{50}$ (4) $\frac{9}{50}$

**Answer (3)**



**Sol.** Take $P(A)$ = Probability that number is multiple of 4

$P(B)$ = Probability that number is multiple of 6

$P(C)$ = Probability that number is multiple of 7

$$P(A)=\frac{12}{50}, P(B)=\frac{8}{50} \text{ and } P(C)=\frac{7}{50}$$

$$P(A\cap B)=\frac{4}{50} \text{ (Multiple of 12)}$$

$$P(B\cap C)=\frac{1}{50} \text{ (Multiple of 42)}$$

$$P(A\cap C)=\frac{1}{50} \text{ (Multiple of 28)}$$

$$P(A\cap B\cap C)=0 \text{ (Multiple of 84)}$$

$$P(A\cup B\cap C)=P(A)+P(B)+P(C)-$$

$$P(A\cap B)-P(B\cap C)-P(A\cap C)+P(A\cap B\cap C)$$

$$=\frac{12}{50}+\frac{8}{50}+\frac{7}{50}-\frac{4}{50}-\frac{1}{50}-\frac{1}{50}+0=\frac{21}{50}$$

9. Let a unit vector $\hat{u}=x\hat{i}+y\hat{j}+z\hat{k}$ make angles $\frac{\pi}{2}, \frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ respectively, If $\vec{v}=\frac{1}{2}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$, then $|\hat{u}-\vec{v}|^2$ is equal to

(1) 9 (2) 7

(3) $\frac{11}{2}$ (4) $\frac{5}{2}$

**Answer (4)**

**Sol.** $$\frac{x}{\sqrt{2}}+\frac{z}{\sqrt{2}}=0 \quad \ldots (1)$$

$$\frac{y}{\sqrt{2}}+\frac{z}{\sqrt{2}}=\frac{1}{2} \quad \ldots (2)$$

$$\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{2}}=\frac{-1}{2} \quad \ldots (3)$$

From (1) and (3)

$$\frac{y}{\sqrt{2}}-\frac{z}{\sqrt{2}}=-\frac{1}{2} \quad \ldots (4)$$

From (2) and (4)

$\sqrt{2}y=0$

$y=0$

and $z=\frac{1}{\sqrt{2}}, x=-\frac{1}{\sqrt{2}}$

$$\therefore \vec{v}-\vec{u}=\sqrt{2}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$$

$$|\vec{v}-\vec{u}|=\sqrt{2+\frac{1}{2}}$$

$$|\vec{v}-\vec{u}|=\sqrt{\frac{5}{2}}$$

$$\therefore |\vec{v}-\vec{u}|^2=\frac{5}{2}$$

10. If each term of a geometric progression $a_1$, $a_2$, $a_3$, ..... with $a_1=\frac{1}{8}$ and $a_2\neq a_1$, is arithmetic mean of the next two terms and $S_n=a_1+a_2+....+a_n$, then $S_{20}-S_{18}$ is equal to

(1) $-2^{18}$ (2) $2^{18}$

(3) $-2^{15}$ (4) $2^{15}$

**Answer (3)**

**Sol.** Given $a_1=\frac{1}{8}$

$a, ar, ar^2, ar^3, ....$

Also, $a_2$ is arithmetic mean of next two terms

$\therefore$ $2ar=ar^2+ar^3$

$2ar=r(ar+ar^2)$

$2a=ar+ar^2$

$2=r+r^2$

$r^2+r-2=0$

$(r+2)(r-1)=0$

$r\neq 1$



$\Rightarrow \quad r = -2$

$\therefore \quad S_{20} - S_{18} = \dfrac{a(1-r^{20})}{1-r} - \dfrac{a(1-r^{18})}{1-r}$

$= \dfrac{1}{8}\left(\dfrac{1}{3}\left(1-r^{20}-1+r^{18}\right)\right)$

$= \dfrac{1}{24}\cdot 2^{18}\left(1-4\right)$

$= \dfrac{-2^{18}}{8} \Rightarrow -2^{15}$

11. Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $\left|P^{-1}AP - 2I\right|$ is equal to

(1) 26 (2) 27

(3) 66 (4) 23

**Answer (1)**

**Sol.** $|P^{-1}\,AP - 2I|$

$= |P^{-1}\,AP - 2\,(P^{-1}\,P)|$

$= |P^{-1}I|\;|AP - 2P|$

$= |P^{-1}I|\;|A - 2I|\;|P|$

$= |A - 2I|$

$A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$

$A - 2I = \begin{bmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{bmatrix}$

$|A - 2I| = -1\,(-33) + 2\,(18)$

$= 69$

Prime factors of 69 = 3, 23

Sum = 26

12. Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to

(1) 15 (2) 18

(3) 12 (4) 16

**Answer (1)**

**Sol.** Given that 8 identical books have to be arranged in 4 identical shelves.

| I | II | III | IV |
|---|---|---|---|
| 8 | 0 | 0 | 0 |
| 7 | 1 | 0 | 0 |
| 6 | 2 | 0 | 0 |
| 6 | 1 | 1 | 0 |
| 5 | 3 | 0 | 0 |
| 5 | 2 | 1 | 0 |
| 5 | 1 | 1 | 1 |
| 4 | 4 | 0 | 0 |
| 4 | 3 | 1 | 0 |
| 4 | 2 | 2 | 0 |
| 4 | 2 | 1 | 1 |
| 3 | 3 | 2 | 0 |
| 3 | 3 | 1 | 1 |
| 3 | 2 | 2 | 1 |
| 2 | 2 | 2 | 2 |

Total 15 cases are possible.

13. Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 12\vec{a} + 4\vec{b}$, where $O$ is the origin, If $S$ is the parallelogram with adjacent sides $OA$ and $OC$, then $\dfrac{\text{area of the quadrilateral } OA\,BC}{\text{area of } S}$ is equal to _____

(1) 6 (2) 7

(3) 10 (4) 8

**Answer (4)**



**Sol.** $S = \left|\overline{OA} \times \overline{OC}\right| = \left|\bar{a} \times \bar{b}\right|$

Area of quadrilateral $OABC$

$= \frac{1}{2}\left|\overline{CA} \times \overline{OB}\right|$

$= \frac{1}{2}\left|\left(\bar{b} - \bar{a}\right) \times \left(12\bar{a} + 4\bar{b}\right)\right|$

$= \frac{1}{2} \cdot 16\left|\bar{a} \times \bar{b}\right|$

Required ratio = 8

14. If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ respectively and the mean of the first four observations is $\frac{7}{2}$, then the variance of the first four observations in equal to

(1) $\frac{77}{12}$ (2) $\frac{105}{4}$

(3) $\frac{4}{5}$ (4) $\frac{5}{4}$

**Answer (4)**

**Sol.** $n = 5$

Mean $\bar{x} = \frac{24}{5}$

Take data as $x_1, x_2, x_3, x_4, x_5$

So, $x_1 + x_2 + x_3 + x_4 + x_5 = 24$

Mean of first 4 observation is $\frac{7}{2}$

$x_1 + x_2 + x_3 + x_4 = 14$

$\Rightarrow x_5 = 10$

Variance of first 5 observations.

$$\frac{\sum_{i=1}^{5} x_i^2}{n} - \left(\bar{x}\right)^2 = \frac{194}{25}$$

$$\frac{\sum_{i=1}^{5} x_i^2}{5} - \frac{576}{25} = \frac{194}{25}$$

$$\Rightarrow 5\sum_{i=1}^{5} x_i^2 - 576 = 194$$

$$\Rightarrow 5\sum_{i=1}^{5} x_i^2 = 770$$

$$\Rightarrow x_1^2 + x_2^2 + x_3^2 + x_4^2 + 100 = 154$$

$$\Rightarrow \sum_{i=1}^{4} x_i^2 = 54$$

Variance of first 4 terms:

$$\frac{\sum_{i=1}^{4} x_i^2}{4} - \frac{49}{4} = \frac{54 - 49}{4} = \frac{5}{4}$$

15. The distance of the point (2, 3) from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to

(1) $4\sqrt{2}$ (2) $6\sqrt{3}$

(3) $3 + 4\sqrt{2}$ (4) $4 + 6\sqrt{3}$

**Answer (4)**

**Sol.** Given line $y = \sqrt{3}x + 1$

Inclination = $\frac{\pi}{3}$

Any point on line though (2, 3) and parallel to given line.

$$P\left(2 + r\cos\frac{\pi}{3}, 3 + r\sin\frac{\pi}{3}\right) \equiv \left(2 + \frac{r}{2}, 3 + \frac{r\sqrt{3}}{2}\right)$$

If $P$ lies on $2x - 3y + 28 = 0$

$$2\left(2 + \frac{r}{2}\right) - 3\left(3 + \frac{r\sqrt{3}}{2}\right) + 28 = 0$$

$\Rightarrow |r| = 4 + 6\sqrt{3}$ = Required distance



16. The function $f(x) = 2x + 3(x)^{\frac{2}{3}}$, $x \in R$, has

(1) Exactly one point of local minima and no point of local maxima

(2) Exactly one point of local maxima and no point of local minima

(3) Exactly two points of local maxima and exactly one point of local minima

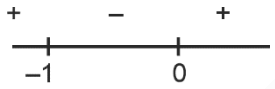
(4) Exactly one point of local maxima and exactly one point of local minima

**Answer (4)**

**Sol.** $f(x) = 2x + 3x^{\frac{2}{3}}, \quad x \in R$

$$f'(x) = 2 + 3\left(\frac{2}{3}\right)x^{\frac{-1}{3}}$$

$$= 2\left(1 + \frac{1}{x^{\frac{1}{3}}}\right) = \frac{2\left(x^{\frac{1}{3}} + 1\right)}{x^{\frac{1}{3}}}$$




$f'(x)$ changes from +ve to –ve at $x = -1$

$\Rightarrow$ point of local maxima

$f'(x)$ changes from –ve to +ve at $x = 0$

$\Rightarrow$ point of local minima

17. Let $y = \log_e\left(\frac{1-x^2}{1+x^2}\right)$, $-1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225(y' - y'')$ is equal to

(1) 742 (2) 746

(3) 732 (4) 736

**Answer (4)**

**Sol.** $y = \ln\left(\frac{1-x^2}{1+x^2}\right), x \in (-1, 1)$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\ln\left(1-x^2\right)\right) - \frac{d}{dx}\left(\ln\left(1+x^2\right)\right)$$

$$= \frac{1}{\left(1-x^2\right)}(-2x) - \frac{(2x)}{\left(1+x^2\right)}$$

$$= -2x\left[\frac{1}{1-x^2} + \frac{1}{1+x^2}\right] = \frac{-4x}{1-x^4}$$

$$y' = \frac{-4x}{1-x^4}$$

$$y'' = \frac{d}{dx}\left(\frac{-4x}{1-x^4}\right) = \frac{\left(1-x^4\right)[-4] - [-4x]\left[-4x^3\right]}{\left(1-x^4\right)^2}$$

$$= (-4)\frac{\left[1-x^4+4x^4\right]}{\left(1-x^4\right)^2}$$

$$= \frac{(-4)\left(1+3x^4\right)}{\left(1-x^4\right)^2}$$

At $x = \frac{1}{2}$

$$y' = \frac{-2}{1-\frac{1}{16}} = \frac{-32}{15},$$

$$y'' = \frac{(-4)\left[1+\frac{3}{16}\right]}{\left(1-\frac{1}{16}\right)^2} = \frac{(16)(-4)(19)}{225}$$

$$225(y' - y'') = 225\left[\frac{-32}{15} + \frac{16 \times 4 \times 19}{225}\right]$$

$= 16 \times 4 \times 19 - 32 \times 15$

$= 32[38 - 15] = 32 \times 23$

$= 736$



18. If $\int \frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx =$

$A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$, where $C$ is the integration constant, then $AB$ is equal to

(1) $2\sec\theta$ (2) $4\,\text{cosec}(2\theta)$

(3) $4\sec\theta$ (4) $8\,\text{cosec}(2\theta)$

**Answer (4)**

**Sol.** $I=\int\frac{\left(\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x\right)dx}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}$

$$I=\int\frac{\sec^{\frac{3}{2}}x\,dx}{\sqrt{\sin(x-\theta)}}+\int\frac{\text{cosec}^{\frac{3}{2}}x\,dx}{\sqrt{\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x\,dx}{\sqrt{\sec x\sin(x-\theta)}}+\int\frac{\text{cosec}^2 x\,dx}{\sqrt{\text{cosec}\,x\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x\,dx}{\sqrt{\tan x\cos\theta-\sin\theta}}+\int\frac{\text{cosec}^2 x\,dx}{\sqrt{\cos\theta-\cot x\sin\theta}}$$

$$=I_1+I_2$$

$$I_1=\int\frac{\sec^2 x\,dx}{\sqrt{\tan x\cos\theta-\sin\theta}}$$

Let $\tan x\cos\theta-\sin\theta=t^2$

$\Rightarrow \sec^2 x\,dx=\frac{2t\,dt}{\cos\theta}$

$$I_1=\int\frac{\frac{2t\,dt}{\cos\theta}}{t}=\frac{2t}{\cos\theta}+C_1=\frac{2}{\cos\theta}\sqrt{\tan x\cot\theta-\sin\theta}+C_1$$

$$I_2=\frac{2}{\sin\theta}\sqrt{\cos\theta-\sin\theta\cot x}+C_2$$

$\Rightarrow I_1+I_2 \Rightarrow B=\frac{2}{\sin\theta}, A=\frac{2}{\cos\theta}$

$$\Rightarrow AB=\frac{4}{\sin\theta\cos\theta}=8\,\text{cosec}2\theta$$

19. If $R$ is the smallest equivalence relation on the set $\{1, 2, 3, 4\}$ such that $\{(1, 2), (1, 3)\} \subset R$, then the number of elements in $R$ is _____

(1) 8 (2) 10

(3) 12 (4) 15

**Answer (2)**

**Sol.** $R$ is smallest equivalence relation

$\Rightarrow$ $R$ must have (1, 1), (2, 2), (3, 3), (4, 4) (For reflexive)

$\Rightarrow$ Now $R$ is {(1, 1), (1, 2), (2, 2), (1, 3), (3, 3), (4, 4)}

$\Rightarrow$ For symmetric relation

(1, 2) must have (2, 1)

(1, 3) must have (3, 1)

$\Rightarrow$ $R$ now is

{(1, 1), (1, 2), (2, 1), (2, 2), (1, 3), (3, 1), (3, 3), (4, 4)}

Since (2, 1) and (1, 3) is in $R$

$\Rightarrow$ (2, 3) must be there

$\Rightarrow$ (3, 2) must be there

$\Rightarrow$ {(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (3, 2), (2, 3), (1, 3), (3, 1)}

$\Rightarrow$ $R$ must have minimum 10 elements.

20. If $\log_e a$, $\log_e b$, $\log_e c$ are in an A.P. and $\log_e a-\log_e 2b$, $\log_e 2b-\log_e 3c$, $\log_e 3c-\log_e a$ are also in an A.P., then $a : b : c$ is equal to

(1) 25 : 10 : 4 (2) 9 : 6 : 4

(3) 16 : 4 : 1 (4) 6 : 3 : 2

**Answer (2)**

**Sol.** $\ln a$, $\ln b$, $\ln c \rightarrow$ A.P.

$\Rightarrow 2\ln b=\ln a+\ln c$

$\Rightarrow b^2=ac$

$\ln a-\ln 2b$, $\ln 2b-\ln 3c$, $\ln 3c-\ln a \rightarrow$ A.P.



$$\Rightarrow 2\ln\left(\frac{2b}{3c}\right)=\ln\left(\frac{a}{2b}\right)+\ln\left(\frac{3c}{a}\right)=\ln\left[\left(\frac{a}{2b}\right)\left(\frac{3c}{a}\right)\right]$$

$$\Rightarrow \frac{4b^2}{9c^2}=\left(\frac{a}{2b}\right)\left(\frac{3c}{a}\right)=\frac{3c}{2b}$$

$$\Rightarrow 8b^3=27c^5$$

$$\Rightarrow 2b=3c$$

$$\frac{9c^2}{4}=ac \Rightarrow c=\frac{4}{9}a$$

$$c=\frac{2}{3}b$$

$$a:b:c=\frac{9}{4}c:\frac{3}{2}c:c$$

$$a:b:c \Rightarrow =\frac{9}{4}:\frac{3}{2}:c \Rightarrow a:b:c \Rightarrow 9:6:4$$

## SECTION - B

**Numerical Value Type Questions:** This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

21. Let the set $C=\left\{(x,y)\,|\,x^2-2^y=2023, x,y\in \mathrm{N}\right\}$. Then $\sum\limits_{(x,y)\in C}(x+y)$ is equal to ____.

**Answer (46)**

**Sol.** $C=\left\{(x,y)\,|\,x^2-2^y=2023, x,y\in N\right\}$

$$x^2-2023=2^y$$

$$x^2-2025=2^y-2$$

$$\Rightarrow x^2-45^2=2^y-2$$

$$= (x-45)(x+45)=2(2^{y-1}-1)$$

For $x\in$ even, no solution

$x\in$ odd, $\Rightarrow x=2m+1$

$$(2m-44)(2m+46)=2(2^{y-1}-1)$$

$$\Rightarrow 4(m-22)(m+23)=2\underbrace{(2^{y-1}-1)}_{\text{Odd Number}}$$

$\Rightarrow$ No solution form

$\Rightarrow x = 45$ or $x = -45$

$\Rightarrow 2^{y-1}=1 \Rightarrow y=1$

(45, 1), (–45, 1) solution

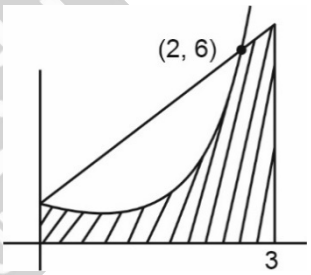
$\because x, y\in N \Rightarrow x+y=45+1=46$

22. Let the area of the region $\{(x, y): 0\le x\le 3, 0\le y\le \min\{x^2+2, 2x+2\}\}$ be $A$. Then $12A$ is equal to _____.

**Answer (164)**

**Sol.** $\{(x, y): 0\le x\le 3, 0\le y\le \min\{x^2+2, 2x+2\}\}$

$$\min(x^2+2, 2x+2)=\begin{cases}x^2+2 & 0\le x\le 2\\ 2x+2 & 2\le x\le 3\end{cases}$$




$$A=\int_0^2(x^2+2)dx+\frac{1}{2}[6+8]\times 1=164$$

23. Let $O$ be the origin, and $M$ and $N$ be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ respectively such that $MN$ is the shortest distance between the given line. Then $\overrightarrow{OM}\cdot\overrightarrow{ON}$ is equal to _____.

**Answer (9)**



**Sol.** $L_1 : \dfrac{x-5}{4} = \dfrac{y-4}{1} = \dfrac{z-5}{3} = \lambda$

Any point of $L_1$ : $(4\lambda + 5,\ \lambda + 4,\ 3\lambda + 5) : M$

$L_2 : \dfrac{x+8}{12} = \dfrac{y+2}{5} = \dfrac{z+11}{9} = \mu$

Any point on $L_2$ : $(12\mu - 8,\ 5\mu - 2,\ 9\mu - 11) : N$

DR of $MN < 4\lambda - 12\mu + 13,\ \lambda - 5\mu + 6,\ 3\lambda - 9\mu + 16 >$

Now

$4(4\lambda - 12\mu + 13) + (\lambda - 5\mu + 6) + 3(3\lambda - 9\mu + 16) = 0$ and

$12(4\lambda - 12\mu + 13) + 5(\lambda - 5\mu + 6) + 9(3\lambda - 9\mu + 16) = 0$

$\Rightarrow \lambda = -1$ and $\mu = 1$

$\therefore$ $M(1, 3, 2)$ and $N(4, 3, -2)$

$OM \cdot ON = 1 \cdot (4) + 3 \cdot (3) + 2(-2)$

$= 4 + 9 - 4 = 9$

24. If $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x}\, dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$, where $\alpha$, $\beta$ and $\gamma$ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____.

**Answer (6)**

**Sol.** $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x}\, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} |\sin x - \cos x|\, dx$

$= \displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\cos x - \sin x)\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sin x - \cos x)\, dx$

$= 2\sqrt{2} - \sqrt{3} - 1$

$\Rightarrow \alpha = -1 \quad \beta = 2, \quad \gamma = -1$

$\therefore\ 3\alpha + 4\beta - \gamma = 3(-1) + 4(2) - (-1)$

$= 6$

25. Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

**Answer (1)**

**Sol.** Remainder $64^{32^{32}}$ when divided by 9.

$64 \equiv 1 \pmod 9$

$64^{32^{32}} \equiv 1^{32^{32}} \pmod 9$

$\Rightarrow$ Remainder = 1

26. Let the slope of the line $45x + 5y + 3 = 0$ be $27r_1 + \dfrac{9r_2}{2}$ for some $r_1, r_2 \in \mathbb{R}$. Then $\lim\limits_{x \to 3} \left( \displaystyle\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x}\, dt \right)$ is equal to _____.

**Answer (12)**

**Sol.** $27r_1 + \dfrac{9r_2}{2} = -9$

$54r_1 + 9r_2 + 18 = 0$

$\ell = \lim\limits_{x \to 3} \left( \displaystyle\int_3^x \frac{8t^2 dt}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} \right)$

$= \lim\limits_{x \to 3} \dfrac{\displaystyle\int_3^x 8t^2 dt}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x}$ Form : $\dfrac{0}{0}$

Using L-H rule

$\lim\limits_{x \to 3} \dfrac{8x^2}{\frac{3r_2}{2} - 2r_2 x - 3r_1 x^2 - 3} = \dfrac{8 \times 9 \times 2}{-9r_2 - 54r_1 - 6} = 12$



27. Let $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\dfrac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$ be differentiable in $(-\infty,0)\cup(0,\infty)$ and $f(1)=1$. Then the value of $ea$, such that $f(a)=0$, is equal to _____.

**Answer (2)**

**Sol.** $f(x)=\sqrt{\lim\limits_{r\to x}\left(\dfrac{2r^2f(r)f(r)-f(x)}{(r+x)(r-x)}-r^3e^{\frac{f(r)}{r}}\right)}$

$$f(x)=\sqrt{\frac{2x^2f(x)f'(x)}{2x}-x^3e^{\frac{f(x)}{x}}}$$

$$\Rightarrow\quad y^2=xy\frac{dy}{dx}-x^3e^{\frac{y}{x}}$$

$$\Rightarrow\quad \frac{y^2}{x^2}=\frac{y}{x}\frac{dy}{dx}-xe^{\frac{y}{x}}$$

Put $y=tx$

$$\Rightarrow\quad \frac{dy}{dx}=t+x\frac{dt}{dx}$$

$$t^2=t^2+tx\frac{dt}{dx}-xe^t$$

$$t\frac{dt}{dx}=e^t$$

$$dx=te^{-t}dt$$

$$\Rightarrow\quad x=-e^{-t}(t+1)+c$$

$$x=-e^{-\frac{y}{x}}\left(\frac{y}{x}+1\right)+c\quad\because f(1)=1\Rightarrow c=1+\frac{2}{e}$$

$f(a)=0)$

$$a=-1(0+1)+1+\frac{2}{e}$$

$$\Rightarrow\quad ae=2$$

28. Let $P(\alpha,\beta)$ be a point on the parabola $y^2=4x$. If $P$ also lies on the chord of the parabola $x^2=8y$ whose mid-point is $\left(1,\dfrac{5}{4}\right)$, then $(\alpha-28)(\beta-8)$ is equal to _____.

**Answer (192)**

**Sol.** $P(\alpha,\beta)$ lies on $y^2=4x\Rightarrow\beta^2=4\alpha$ ...(i)

Equation of chord of $x^2=8y$ whose mid-point is $\left(1,\dfrac{5}{4}\right)$ is

$T=S_1$

$$x\times1-4\left(y+\frac{5}{4}\right)=1^2-8\cdot\frac{5}{4}$$

$x-4y-5=-9$

$x-4y+4=0$

$\because$ $(\alpha,\beta)$ lies on $x-4y+4=0$

$\Rightarrow\quad \alpha-4\beta+4=0$ ...(ii)

$$\frac{\beta^2}{4}-4\beta+4=0$$

$\beta^2-16\beta+16=0$

$(\beta-8)^2=48\Rightarrow\beta=8\pm4\sqrt{3}$

$\because$ Intersection point of $x^2=8y$ and $x-4y+4=0$ are $\left(-2,\dfrac{1}{2}\right)$ and $(4,2)$

$$\Rightarrow\quad \alpha\in(-2,4),\beta\in\left(\frac{1}{2},2\right)$$

$$\Rightarrow\quad \beta=8-4\sqrt{3}\Rightarrow(\beta-8)=-4\sqrt{3}$$

$$\Rightarrow\quad \alpha=32-16\sqrt{3}-4$$

$$\alpha=28-16\sqrt{3}\Rightarrow(\alpha-28)=-16\sqrt{3}$$

$\Rightarrow\quad (\alpha-28)(\beta-8)=192$



29. Let $\alpha$, $\beta$ be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that Im($\alpha$) > Im($\beta$). Let $a$, $b$ be integers not divisible by 3 and $n$ be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

**Answer (49)**

**Sol.** $x^2 - \sqrt{6}x + 3 = 0$

$$x = \frac{\sqrt{6} \pm \sqrt{6-12}}{2}, \quad \alpha + \beta = \sqrt{6}, \quad \alpha\beta = 3$$

$$x = \frac{\sqrt{6}}{2} \pm \frac{\sqrt{6}}{2}i$$

$$\alpha = \frac{\sqrt{6}}{2}(1+i), \quad \beta = \frac{\sqrt{6}}{2}(1-i)$$

$$S = \frac{\alpha^{99}}{\beta} + \alpha^{98}$$

$$\Rightarrow \quad = \alpha^{99}\left(\frac{1}{\beta} + \frac{1}{\alpha}\right)$$

$$= \alpha^{99}\left(\frac{\alpha+\beta}{\alpha\beta}\right)$$

$$= \alpha^{99}\left(\frac{\sqrt{6}}{3}\right)$$

$$= \frac{\left(\sqrt{6}\right)^{100}}{2^{99}\cdot 3} \times (1+i)^{99}$$

$\because$ $(1+i) = \sqrt{2}e^{i\frac{\pi}{4}}$

$$\Rightarrow (1+i)^{99} = \left(\sqrt{2}\right)^{99} e^{i\frac{99\pi}{4}}$$

$$= \left(\sqrt{2}\right)^{99}\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right)$$

$$\Rightarrow \quad S = \frac{\left(\sqrt{6}\right)^{100}}{3\cdot 2^{99}} \times \left(\sqrt{2}\right)^{98}(1-i)$$

$$= \frac{\left(\sqrt{3}\right)^{100}}{3}(1-i) = 3^{49}(1-i)$$

$\Rightarrow$ $n = 49$, $a = 1$, $b = -1$

$\Rightarrow$ $n + a + b = 49$

30. Let for any three distinct consecutive terms $a$, $b$, $c$ of an A.P., the lines $ax + by + c = 0$ be concurrent at the point $P$ and $Q(\alpha, \beta)$ be a point such that the system of equations.

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ and

$x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____.

**Answer (113)**

**Sol.** $b - a = c - b$

$a - 2b + c = 0$

$\Rightarrow$ $ax + by + c = 0$ are concurrent at $P(1, -2)$

$\because$ $x + y + z = 6$,

$2x + 5y + \alpha z = \beta$,

$x + 2y + 3z = 4$

$\Delta = 8 - \alpha$,

$\Delta_1 = -8\alpha - \beta + 70$,

$\Delta_2 = 2\alpha + 2\beta - 28$,

$\Delta_3 = 6 - \beta$ for infinitely many solutions

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Rightarrow$ $\alpha = 8$, $\beta = 6$

$\Rightarrow$ $Q(8, 6)$, $P(1, -2)$

$\Rightarrow$ $(PQ)^2 = 113$



# PHYSICS

## SECTION - A

**Multiple Choice Questions:** This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

**Choose the correct answer :**

31. A stone of mass 900 g is tied to a string and moved in a vertical circle of radius 1 m making 10 rpm. The tension in the string, when the stone is at the lowest point is

(if $\pi^2 = 9.8$ and $g = 9.8$ m/s$^2$)

(1) 9.8 N (2) 8.82 N

(3) 97 N (4) 17.8 N

**Answer (1)**

**Sol.** $\omega = 10 \text{ rpm} = 10 \times \frac{\pi}{30} = \frac{\pi}{3}$ rad/s

$T - mg = m\omega^2 r$ $\quad m = 0.9$ kg, $r = 1$ m

$T - 0.9 \times 9.8 = 0.9 \times \frac{\pi^2}{9} \times 1$

$T = 0.9 \times 9.8 \left\{1 + \frac{1}{9}\right\}$

$= 0.9 \times 9.8 \times \frac{10}{9}$

$= 9.8$ N

32. The temperature of a gas having $2.0 \times 10^{25}$ molecules per cubic meter at 1.38 atm

(Given, $k = 1.38 \times 10^{-23}$ JK$^{-1}$) is

(1) 300 K (2) 200 K

(3) 500 K (4) 100 K

**Answer (3)**

**Sol.** $PV = NkT$

$\Rightarrow P = \frac{N}{V}kT = nkT$

$\Rightarrow 1.38 \times 101325 = 2 \times 10^{25} \times 1.38 \times 10^{-23} T$

$\Rightarrow T = \frac{101325}{200} \approx 500$ K

33. If the distance between object and its two times magnified virtual image produced by a curved mirror is 15 cm, the focal length of the mirror must be

(1) –12 cm

(2) $\frac{10}{3}$ cm

(3) –10 cm

(4) 15 cm

**Answer (3)**

**Sol.** Let $|u| = x$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\quad m = -\frac{v}{u}$

$\frac{1}{2x} - \frac{1}{x} = \frac{1}{f}$ $\quad 2 = \frac{-v}{-x}$

$-\frac{1}{2x} = \frac{1}{f}$ $\quad v = 2x$

$f = -2x$ $\quad$ Also $3x = 15 \Rightarrow x = 5$ cm

$f = -10$ cm

